Secure By Design

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COURSE 1.19 / COMPANION WORKBOOK

Prove you can recover

Backups with custody, one rehearsed restore & the closing record

Close the Build path by proving the network can come back: scheduled directory backups and a CA backup under ceremony-grade custody, Group Policy, DHCP, the repository and analysis configuration, the log archive, and the living records in one hashed catalog, a timed restore rehearsal in isolation, and the recovery record that completes the Body of Evidence.

MATERIALS EDITION 1.4 / COURSE 1.19

Updated 14 October 2026 · sbd-prove-recovery

NAME _____ / DATE _____

START HERE / YOUR LEARNING PATH

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Course review and sign-off

> **HOW TO USE IT** Each lesson page has the same shape: what you will learn, the build steps, the commands that verify them, and a record table. Fill the table from your own environment — never from the sample values.

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Before you start

The model's final chapter: a network that cannot come back was never really built. This course backs up what matters — the directory, the issuing CA, Group Policy, DHCP, the repository and analysis configuration, the log record, and the living records — proves one real restore in isolation, and files the tested-recovery evidence the authorization path expects.

Prerequisites

- ☐ Complete courses 1.1–1.18 (courses 1.16 and 1.17 are coming soon and aren't needed): the full network, its records, the protected log archive, and the living inventory all exist.
- ☐ Provision a protected backup volume (E: in this sample) on both domain controllers and SBD-SUBCA01, written only by the backup job and never one of the volumes being backed up; the member servers keep the backup sets their own courses made, such as SBD-DHCP01's D:\Backups\DHCP. Also provide an offline backup cabinet with removable media and a custody register, and an isolated recovery network segment (or host) for the restore rehearsal.
- ☐ Have the custody records handy: the sealed Directory Services Restore Mode (DSRM) passwords set in course 1.1 and the course 1.6 ceremony record for SBD-ROOTCA01's protected backup — recovery is where those promises come due.

What this course builds

- Back up the directory, CA, policy, DHCP, repository, analysis configuration, logs, and records into one verified catalog.
- Rehearse a real system state restore in isolation, measured against written objectives.
- File the tested-recovery record that completes the Body of Evidence for path 2.

Systems in scope

RECORD	YOUR OBSERVATION
SBD-DC01 — name, address, OS build	
SBD-DC02 — name, address, OS build	
SBD-SUBCA01 — name, address, OS build	
SBD-DHCP01 — name, address, OS build	
SBD-AUDIT01 — name, address, OS build	
SBD-REPO01 — name, address, OS build	

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RECORD	YOUR OBSERVATION
SBD-SPLUNK01 — name, address, OS build	
SBD-MGMT01 — name, address, OS build	

LESSON 01 / ABOUT 20 MIN

Decide what recovery must prove

> **WHY THIS MATTERS** Recovery starts as a list of promises: which systems and records must be restorable, how old a restored copy may be, how long restoration may take — written as commitments the rest of the course exists to keep.

In this lesson you will

- Name the recovery set: what must come back, in what order.
- Set the recovery point and time objectives per item.
- Decide where backups live and who can touch them.

Follow the build

1. Write the recovery set in dependency order — the order the path built it: the directory on both controllers first, because everything trusts it; then the certificate authorities (SBD-SUBCA01's database and key, and SBD-ROOTCA01's course 1.6 ceremony backup); Group Policy; SBD-DHCP01's backup set from course 1.7; the boundary's records (the service-path register and the core router's configuration); SBD-REPO01's snapshot catalog and snapshots (course 1.9); SBD-SPLUNK01's configuration (course 1.12); the log archive and its catalog on SBD-AUDIT01 (course 1.11); and the living records — the inventory on SBD-MGMT01 (course 1.15) and the exception record.
2. Set the two numbers per item: how much history may be lost (recovery point) and how long restoration may take (recovery time) — lab-honest values you will actually measure against in lesson 5, not aspirations. A directory backup older than the forest's tombstone lifetime (180 days by default) can't be restored at all, so the directory's recovery point stays well inside it.
3. Decide backup placement: each protected backup volume is written only by its backup process and readable by the recovery role, and custodians of the matching tier carry copies to the offline backup cabinet on media — the air-gapped network's version of an off-site copy. There is no backup server and no network path to the cabinet.
4. Confirm the standing custody items: the DSRM passwords set in course 1.1 are sealed and findable, and SBD-ROOTCA01's protected backup from the course 1.6 ceremony is accounted for — recovery depends on secrets that must exist before the bad day.

Check your result

```
# On each controller and SBD-SUBCA01: capture the current protection state before this
course changes it.
wbadmin get versions
Get-Volume -DriveLetter E | Select-Object DriveLetter, FileSystemLabel, SizeRemaining   #
the protected backup volume
# The tombstone lifetime bounds the directory's recovery point (empty means the 180-day
default).
(Get-ADObject "CN=Directory Service,CN=Windows NT,CN=Services,$((Get-
ADRootDSE).configurationNamingContext)" -Properties tombstoneLifetime).tombstoneLifetime
```

> **WHAT SUCCESS LOOKS LIKE** The recovery set is ordered by dependency with point and time objectives per item, the backup placement and access rules are written, and the DSRM and root-key custody records check out.

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Your record

Evidence to keep: Keep the recovery requirements sheet, the objectives table, the placement and access decisions, and the custody verification notes.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

Why does the directory lead the recovery order?

REMEMBER The recovery set is ordered by dependency, directory first. Point and time objectives exist to be measured, not admired. Backups get a protected home plus one offline, custody-tracked copy.

Microsoft · Active Directory forest recovery guide

NIST · SP 800-53 Rev. 5 security controls (CP family)

LESSON 02 / ABOUT 25 MIN

Back up the directory, on schedule

WHY THIS MATTERS System state from both controllers, on a schedule, to the protected target — plus the quiet feature that handles the most common 'disaster' of all: the Recycle Bin for deleted directory objects.

In this lesson you will

- Schedule system state and critical-volume backups on both domain controllers.
- Verify and catalog each backup with hashes.
- Enable the Active Directory Recycle Bin for object-level recovery.

Follow the build

1. In a Tier 0 session from AdminJumpBox01, schedule Windows Server Backup on SBD-DC01 and SBD-DC02, each to its own protected backup volume, with the system state and all critical volumes (wbadmin's -systemState and -allCritical): the system state is what a directory restore in DSRM needs, and the critical volumes let the same backup rebuild a controller whose hardware is gone. Two controllers, two independent backups — replication is availability and was never recovery, as course 1.1 put it.
2. After the first run, verify each backup's completion, record its version identifier in the backup catalog (C:\Evidence\recovery on AdminJumpBox01), and hash the catalog — an unverified backup is a hope with a timestamp.
3. Enable the Active Directory Recycle Bin from the Tier 0 seat — a one-way, forest-level change run against the domain naming master (SBD-DC01) and recorded as the change it is — so a deleted user or OU comes back intact without touching system state. Objects deleted before it was enabled can come back only from a backup.
4. Write the restore decision tree: deleted object → Recycle Bin; corrupted DC on its own hardware → system state restore in DSRM (lesson 5's rehearsal); DC whose hardware is gone → full server (bare-metal) recovery from the critical-volumes backup; whole-forest loss → the Microsoft forest recovery guide, referenced and staged with the evidence rather than improvised on the day.

Check your result

```
# Tier 0 session from AdminJumpBox01: a daily schedule on each controller (E: is never a
volume being backed up)
Invoke-Command SBD-DC01.sbd.test, SBD-DC02.sbd.test { wbadmin enable backup -addtarget:E:
-schedule:21:00 -systemState -allCritical -quiet }
Invoke-Command SBD-DC01.sbd.test, SBD-DC02.sbd.test { wbadmin get versions }   # after the
first run
Enable-ADOptionalFeature 'Recycle Bin Feature' -Scope ForestOrConfigurationSet -Target
sbd.test -Server SBD-DC01.sbd.test
Get-ADOptionalFeature -Filter {Name -like 'Recycle Bin*'} | Select-Object Name,
EnabledScopes
Get-FileHash -Algorithm SHA256 C:\Evidence\recovery\backup-catalog.txt
```

WHAT SUCCESS LOOKS LIKE Both DCs show scheduled, completed system state and critical-volume versions on their protected volumes, the Recycle Bin reports an enabled scope, and the decision tree names the tool for each failure class.

Your record

Evidence to keep: Keep the backup schedules and version listings, the hashed catalog, the Recycle Bin enablement record, and the restore decision tree.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

Both DCs replicate — why back up both?

> **REMEMBER** Two independent system state schedules, verified and cataloged. The Recycle Bin handles the everyday disaster in seconds. A decision tree matches each failure class to its tool.

Microsoft · Active Directory forest recovery guide

Microsoft · Wbadmin (Windows Server Backup)

Microsoft · wbadmin enable backup

Microsoft · AD forest recovery: perform a full server recovery

Microsoft · Enable-ADOptionalFeature (Active Directory Recycle Bin)

LESSON 03 / ABOUT 30 MIN

Back up the CA, services, policy, and records

> **WHY THIS MATTERS** The directory is not the only thing the network cannot re-derive: the issuing CA's database and key, the Group Policy set, SBD-DHCP01's database, SBD-REPO01's snapshots, SBD-SPLUNK01's configuration, the log archive, the inventory, and the boundary's records each get their protected, cataloged copy.

In this lesson you will

- Back up the issuing CA's database, key, and configuration to its protected backup volume under ceremony-grade protection.
- Refresh the Group Policy and DHCP backups, with the restore notes each needs.
- Fold the repository, analysis configuration, log archive, inventory, and boundary records into the same catalog.

Follow the build

1. Back up the issuing CA — database and private key — with certutil to SBD-SUBCA01's protected backup volume (E:\Backups\CA in this sample), and export its registry configuration beside it, because a restored database needs the CA's settings too. Never write it to C:\Evidence: the private key is a secret, and secrets never go into evidence. Protect it to the standard of the course 1.6 ceremony, record custody, and put only the hashes and the custody record in evidence; SBD-ROOTCA01's ceremony backup is verified against its record, never duplicated or brought onto the network.
2. Refresh the Group Policy backup the way course 1.5 made it: Backup-GPO -All, now the whole policy estate (course 1.6's SBD Certificate Autoenrollment and course 1.8's tier logon GPOs included), into C:\Evidence\recovery\gpo-backup on AdminJumpBox01, with every link exported beside it — a restored GPO brings back its settings and permissions, not its links. The sealed course 1.5 set stays as it is.
3. Take a fresh SBD-DHCP01 set the course 1.7 way — Backup-DhcpServer for the database and Export-DhcpServer for the configuration and leases, in D:\Backups\DHCP with their hashes — and write its restore note: after Restore-DhcpServer, set the DNS update credential again with Set-DhcpServerDnsCredential from the sealed record, because the backup doesn't carry it.
4. Add the Tier 1 sets that can't be re-derived: SBD-REPO01's snapshot catalog in /srv/repo-records with the current and previous snapshots and their manifests (course 1.9), SBD-SPLUNK01's configuration in /opt/splunk/etc (course 1.12 — it holds keys, so it stays on protected media; the indexes are a working copy, and the record itself lives on SBD-AUDIT01), and D:\Inventory on SBD-MGMT01 (course 1.15). Record their hashes in C:\Evidence\recovery on SBD-ADMIN-SRV01.
5. Confirm the log archive's protection already satisfies this course — SBD-AUDIT01's hashed archives in D:\LogArchive and their catalog (D:\LogArchive\Catalog\catalog.csv) are themselves the record — and add both to the recovery set, so the evidence survives the systems it describes.
6. Export the boundary and living records into the set: the service-path register, the exception record (C:\Evidence\exception-record.csv), this course's own sheets, and a read-only export of the core router's configuration taken with the router's own export function from AdminJumpBox01 — this course reads the router's configuration and changes nothing on it. A recovered network without its records would pass no assessment.

Check your result

```
# On SBD-SUBCA01 (Tier 0 session): the CA to its protected volume, never to the evidence
folder
certutil -backup E:\Backups\CA   # prompts for the password that protects the exported key
reg export HKLM\SYSTEM\CurrentControlSet\Services\CertSvc\Configuration E:\Backups\CA\
certsvc-config.reg
Get-ChildItem E:\Backups\CA -Recurse -File | Get-FileHash -Algorithm SHA256 | Export-Csv
E:\Backups\ca-backup-hashes.csv -NoTypeInformation   # only this list goes to C:\Evidence\
recovery
# On AdminJumpBox01: the whole policy estate and every link
New-Item -ItemType Directory C:\Evidence\recovery\gpo-backup -Force | Out-Null
Backup-GPO -All -Path C:\Evidence\recovery\gpo-backup -Comment 'Course 1.19'
@('DC=sbd,DC=test') + (Get-ADOrganizationalUnit -Filter *).DistinguishedName | ForEach-
Object { (Get-GPInheritance -Target $_).GpoLinks } | Select-Object Target, Order,
DisplayName, Enabled, Enforced | Export-Csv C:\Evidence\recovery\gpo-links.csv -
NoTypeInformation
# On SBD-DHCP01 (Tier 1 session): a fresh set; the restore note adds Set-
DhcpServerDnsCredential
Backup-DhcpServer -Path D:\Backups\DHCP\course-1.19\database
Export-DhcpServer -File D:\Backups\DHCP\course-1.19\SBD-DHCP01.xml -Leases
# On SBD-REPO01 and SBD-SPLUNK01 (Tier 1 SSH sessions): hash what the recovery set carries
sha256sum /srv/repo-records/* > /srv/repo-records-hashes.txt
sudo tar -czf /root/splunk-etc-$(date +%F).tar.gz -C /opt/splunk etc && sudo sha256sum
/root/splunk-etc-*.tar.gz
```

> **WHAT SUCCESS LOOKS LIKE** The CA backup and its configuration sit on SBD-SUBCA01's protected backup volume with key custody recorded and only hashes in evidence, the policy and DHCP sets are current with their restore notes, and the catalog covers the repository, analysis configuration, log archive, inventory, router export, and living records.

Your record

Evidence to keep: Keep the CA backup's hash list and custody record (never the backup itself), the policy backup with its link export, the DHCP set's hashes and restore note, the Tier 1 hash lists, and the complete recovery catalog.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

Why does the CA key backup get ceremony-grade handling?

> **REMEMBER** CA database, key, and configuration: backed up to the protected volume, custody recorded, only hashes in evidence. Policy, DHCP, repository, analysis, logs, inventory, and the boundary's records all live in one hashed catalog. The evidence now survives the systems it describes.

Microsoft · Wbadmin (Windows Server Backup)

Microsoft · Certutil reference (CA backup and restore)

Microsoft · Backup-GPO

Microsoft · Backup-DhcpServer

Microsoft · Set-DhcpServerDnsCredential

LESSON 04 / ABOUT 20 MIN

Carry the copies to the offline cabinet

> **WHY THIS MATTERS** A backup that sits on the server it protects fails with it. Move a copy of each backup to the offline cabinet by hand, under the same custody rules as any media crossing a boundary, and keep the custody record that proves it.

In this lesson you will

- Define which custodian carries which backup, by tier.
- Transfer the copies to offline media with hashes and custody records.
- Prove a copy can be found, verified, and returned for a restore.

Follow the build

1. Assign custodians by tier: a Tier 0 custodian handles the controllers' system state, the issuing CA's database and key backup, the Group Policy set, and the router configuration export; the audit custodian (SBD Audit Custodians, course 1.8) handles the log archive and its catalog from SBD-AUDIT01; a Tier 1 server custodian handles the member-server sets — SBD-DHCP01, SBD-REPO01, SBD-SPLUNK01, and SBD-MGMT01. No one carries a tier they do not hold.
2. For each copy, write the media ID, hash the backup on its protected volume, copy it to encrypted removable media, and verify the hash on the media before it leaves the room. Keep Windows Server Backup's WindowsImageBackup folder layout on the media so wbadmin can read it there.
3. Carry the media to the offline backup cabinet and record the handoff in the custody register: who, what, when, the hashes, and the storage slot. This is the course 1.4 media procedure used for backups.
4. Pull one copy back as a test: find it from the register, verify its hash, and confirm it is readable. Lesson 5's rehearsal restores from a copy that has made this round trip.

Check your result

```
# On SBD-DC02 (Tier 0), with encrypted media MEDIA-0012 attached as F:: hash on the
volume, copy, hash on the media
Get-ChildItem E:\WindowsImageBackup\SBD-DC02 -Recurse -File | Get-FileHash -Algorithm
SHA256 | Export-Csv F:\MEDIA-0012-volume.csv -NoTypeInformation
Copy-Item E:\WindowsImageBackup -Destination F:\ -Recurse
Get-ChildItem F:\WindowsImageBackup\SBD-DC02 -Recurse -File | Get-FileHash -Algorithm
SHA256 | Export-Csv F:\MEDIA-0012-media.csv -NoTypeInformation
Compare-Object (Import-Csv F:\MEDIA-0012-volume.csv).Hash (Import-Csv F:\MEDIA-0012-
media.csv).Hash   # no output means the copy matches
wbadmin get versions -backupTarget:F: -machine:SBD-DC02   # the media is readable as a
backup
```

> **WHAT SUCCESS LOOKS LIKE** Every item in the recovery set has an offline copy with matching hashes on the volume and the media, a custody entry with a named custodian of the right tier, and one copy has been retrieved and verified.

Your record

Evidence to keep: Keep the custodian assignments, the hash pairs (copied into C:\Evidence\recovery on the tier's seat), the custody register entries, and the retrieval test record.

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RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

Who carries the domain controllers' system-state backup to the cabinet?

> **REMEMBER** Custodians carry only the backups their tier allows. Every copy has matching hashes and a custody entry. A copy has been retrieved and verified before anyone needs it.

Microsoft · Wbadmin (Windows Server Backup)

NIST · SP 800-53 Rev. 5 security controls (CP family)

LESSON 05 / ABOUT 35 MIN

Rehearse a real restore, in isolation

> **WHY THIS MATTERS** One restore, actually performed: a domain controller's system state brought back on the isolated segment, timed against the objective, verified against the directory — plus the small restore everyone will someday need, from the Recycle Bin.

In this lesson you will

- Restore a DC's system state in Directory Services Restore Mode on the isolated recovery segment.
- Measure the restore against the time objective and verify directory health.
- Perform and record an object-level Recycle Bin restore.

Follow the build

1. Stage the rehearsal: the isolated recovery segment has no path to the production VLANs — a restored controller believing it is authoritative must never meet the live directory; that isolation rule is written before the restore starts. Turn the hypervisor's time synchronization off on the rehearsal copy, as for every virtual machine in the path.
2. Restore SBD-DC02 from the offline copy retrieved in lesson 4, timing the work from decision to directory services running against the lesson 1 objective. A system state restore needs the same server instance, so use a copy of SBD-DC02's virtual machine attached only to the isolated segment, start it in DSRM (bcdedit /set safeboot dsrepair, restart, sign in with the DSRM password), and run wbadmin start systemstaterecovery with -authsysvol: the copy is the only controller in its isolated world, so its SYSVOL must be authoritative or SYSVOL and NETLOGON never come up. Restoring to different hardware needs a full server (bare-metal) recovery from the critical-volumes backup instead.
3. Remove the DSRM boot setting (bcdedit /deletevalue safeboot), restart normally, and verify the restored directory the course 1.1 way — services, the SYSVOL and NETLOGON shares, a directory query. Then check the clock: the copy comes back with the time it was saved with, and with no SBD-DC01 to follow nothing corrects it, so compare it with the reference and record the offset; a restored SBD-DC01 in production is checked against the reference before any member follows it (course 1.3). Record what the rehearsal taught — surprises, the measured time, decision-tree updates — then tear the rehearsal down completely.
4. Back in production, from AdminJumpBox01, run the everyday restore: delete a disposable test object, bring it back from the Recycle Bin with attributes intact, and file both rehearsals — the big one and the small one — as the tested-recovery evidence.

Check your result

```
# On the isolated copy of SBD-DC02: start in DSRM, then sign in with the DSRM password.
bcdedit /set safeboot dsrepair
shutdown /r /t 0
wbadmin get versions -backupTarget:F: -machine:SBD-DC02   # F: = the retrieved media
wbadmin start systemstaterecovery -version:<version-id> -backupTarget:F: -machine:SBD-DC02
-authsysvol
bcdedit /deletevalue safeboot   # then restart normally
# After the restart, on the restored copy
Get-Service NTDS, DNS, Netlogon, DFSR | Select-Object Name, Status
Get-SmbShare -Name SYSVOL, NETLOGON
Get-ADDomain -Server localhost | Select-Object DNSRoot, PDCEmulator
```

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```
w32tm /query /status   # then compare Get-Date with the reference clock and record the
offset
# Back in production, from AdminJumpBox01: the Recycle Bin rehearsal
Get-ADObject -Filter "Name -eq '<test object name>'" -IncludeDeletedObjects | Restore-
ADObject
```

WHAT SUCCESS LOOKS LIKE The isolated restore ran in DSRM and reached a healthy directory within (or honestly against) the time objective, its clock offset is recorded, the DSRM boot setting is removed, the teardown is confirmed, and the production object restore returned the test object intact.

Your record

Evidence to keep: Keep the timed rehearsal log with its verification output and clock reading, the lessons-learned and decision-tree updates, the teardown confirmation, and the object-restore record.

RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

Why must the rehearsal segment be isolated from production?

REMEMBER One real restore, isolated, timed, verified, and torn down. The measured number replaced the hoped-for one. The everyday restore is proven alongside the dramatic one.

Microsoft · Active Directory forest recovery guide

Microsoft · wbadmin start systemstaterecovery

Microsoft · AD forest recovery: perform a nonauthoritative restore

Microsoft · Restore a virtual domain controller

Microsoft · AD forest recovery: perform a full server recovery

Microsoft · Windows Time service tools and settings

LESSON 06 / ABOUT 20 MIN

File the recovery record and close the path

> **WHY THIS MATTERS** The Build path ends the way it worked: owners, schedules, and evidence. The recovery record joins the living documents, the closing SCAP results snapshot the finished network, and the Body of Evidence is ready for the authorization courses to assemble.

In this lesson you will

- Assemble the tested-recovery record with owners and review dates.
- Take the closing SCAP snapshot across the finished network.
- Hand the complete evidence set to the authorization path.

Follow the build

1. Assemble the recovery record: the requirements and objectives, the backup catalog, both rehearsal results with the measured times, the decision tree, the DHCP restore note, and a named owner with a scheduled next rehearsal — recovery is a practice with a calendar, not a binder. Keep its Tier 0 half in C:\Evidence\recovery on AdminJumpBox01 and its Tier 1 half on SBD-ADMIN-SRV01, each sealed with a manifest.
2. Take the closing snapshot: current Security Content Automation Protocol (SCAP) results for every system on the course 1.5 cadence, each tier from its own seat — Tier 0 from AdminJumpBox01, the Windows and Linux member servers from SBD-ADMIN-SRV01, the fleet from SBD-ADMIN-PC01 — with the recorded benchmark versions, triaged as always: the dated 'as-authorized' picture path 2 will reference. Credentialed vulnerability scanning joins it when course 1.17 (coming soon) arrives.
3. Reconcile the three living records one final time — the inventory on SBD-MGMT01 against the directory, DHCP leases, and WSUS rings (the course 1.15 method), the register against observed behavior, and the exception record against open findings — so the authorization path inherits agreement, not archaeology.
4. Write the Build path's closing note: what exists, where every record lives, and what is intentionally open — then start course 2.1, where this network stops being a project and becomes a system seeking authorization.

Check your result

```
# On AdminJumpBox01 (Tier 0), then the same on SBD-ADMIN-SRV01 (Tier 1): seal each half of the recovery record
Get-ChildItem C:\Evidence -Directory | Select-Object Name
Get-ChildItem C:\Evidence\recovery -Recurse -File | Get-FileHash -Algorithm SHA256 |
Export-Csv C:\Evidence\recovery-manifest.csv -NoTypeInformation
Get-FileHash -Algorithm SHA256 C:\Evidence\recovery-manifest.csv
```

> **WHAT SUCCESS LOOKS LIKE** The recovery record is complete with an owner and next-rehearsal date and sealed on both seats, the closing SCAP snapshot is filed with triage, and all three living records reconcile cleanly into the handoff note.

Your record

Evidence to keep: Finish the checklist: the recovery record and its two manifests, the closing SCAP snapshot, the three reconciliations, and the Build path's closing note.

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RECORD	YOUR OBSERVATION
Host(s) and time collected	
Observed result	
Differences from the plan	
Evidence file names	
Open items or findings	

Think it through

What makes recovery 'a practice with a calendar'?

> **REMEMBER** Recovery has an owner, a date, and a measured track record. The closing snapshot freezes the as-authorized picture. The Body of Evidence exists — path 2 assembles, it doesn't excavate.

Microsoft · Active Directory forest recovery guide

NIST · SP 800-53 Rev. 5 security controls (CP family)

Course review and sign-off

Before moving on, confirm each lesson left evidence behind — and that what remains open is written down, not remembered.

Lesson completion

LESSON	EVIDENCE FILED	DATE
01 Decide what recovery must prove	☐ yes	
02 Back up the directory, on schedule	☐ yes	
03 Back up the CA, services, policy, and records	☐ yes	
04 Carry the copies to the offline cabinet	☐ yes	
05 Rehearse a real restore, in isolation	☐ yes	
06 File the recovery record and close the path	☐ yes	

Open items carried forward

Next in the learning path

That completes the Build the Network path. Continue with course 2.1 to carry this network — and its evidence — through the Risk Management Framework to an authorization decision.

> **KEEP IT HONEST** Lesson completion records learning activity; it is not a system authorization decision. Your observations, not the sample values, establish what happened in your environment.

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